We shall deal with positive form as linear operator at first, and we will generalize.

Def. A linear operator $T: V \to V$ on a fin. dim inn. prod space V is called positive if:

T is Hermitian and $\langle T(\alpha), \alpha \rangle > 0$ for all $\alpha \neq 0$

Prop. Let T, U be Hermitian on fin. dim inn. prod space V.
And, $A = [T]_\beta$, where β is any orthonormal basis for V.
1). T is positive if and only if all eigenvalues of T are positive.
2). T is positive if and only if $\sum_{i,j} A_{ij} a_j \overline{a_i} > 0$ for all nonzero tuple $(a_1, \cdots, a_n)$.
3). T is positive if and only if

$$A = B^* B \quad \text{for some } B.$$

Proof. Denote $\beta = \{\alpha_1, \cdots, \alpha_n\}$. Let $B = [I]_\beta^{\{e_1, \cdots, e_n\}} = \begin{pmatrix} \alpha_1 & \alpha_2 & \cdots & \alpha_n \end{pmatrix}$.
And, let $\beta' = \{\beta_1, \cdots, \beta_n\}$ be an orthonormal basis for V consisting eigenvectors of T.

1). If T is positive, then $\langle T(\beta_i), \beta_i \rangle = \langle c_i \beta_i, \beta_i \rangle = c_i > 0$, where c_i is eigenvalue.
Conversely, given $X \in V$, $X = x_1 \beta_1 + \cdots + x_n \beta_n = \langle X, \beta_1 \rangle \cdot \beta_1 + \cdots + \langle X, \beta_n \rangle \cdot \beta_n$, so

$$\langle T(X), X \rangle = \left\langle \langle X, \beta_1 \rangle \cdot c_1 \cdot \beta_1 + \cdots + \langle X, \beta_n \rangle c_n \beta_n, \langle X, \beta_1 \rangle \cdot \beta_1 + \cdots + \langle X, \beta_n \rangle \cdot \beta_n \right\rangle$$

$$= \sum_{i=1}^n c_i \cdot |\langle X, \beta_i \rangle|^2 > 0 \quad \text{if } X \neq 0 \text{ and } c_i > 0.$$

2). Given $X \in V$ s.t. $[X]_\beta = (a_1, \cdots, a_n)$,

$$\langle T(X), X \rangle = \left\langle \sum_{j=1}^n a_j T(\alpha_j), \sum_{i=1}^n a_i \cdot \alpha_i \right\rangle = \sum_{j=1}^n \sum_{i=1}^n \langle T(\alpha_j), \alpha_i \rangle \cdot a_j \cdot \overline{a_i},$$ So proved.

(+Recall: $A_{ij} = \langle T(\alpha_j), \alpha_i \rangle$).

3). If $A = B^* B$, then $\langle B^* B x, x \rangle = \langle Bx, Bx \rangle = \|Bx\|^2 > 0$. Conversely,
since $[T]_{\beta'} = D$ is diagonal with positive values, where

$$D = \begin{pmatrix} c_1 & & 0 \\ & c_2 & \\ 0 & & \ddots \; c_n \end{pmatrix} \quad c_i\text{'s are eigenvalues of } T$$

so we can define $\sqrt{D} = \begin{pmatrix} \sqrt{c_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{c_n} \end{pmatrix}$. Now, for $P = [I]_{\beta'}^{\beta}$,

$$A = P^{-1} D P = P^{-1} \cdot \sqrt{D} \cdot \sqrt{D} \cdot P = (\sqrt{D} P)^* \cdot (\sqrt{D} P),$$ since P is unitary, $\sqrt{D}$ is symmetric.

And, there are more interesting problems,

4). If T and U are positive s.t $T^2 = U^2$, then $T = U$.

5). If T and U are positive s.t $TU = UT$, then TU is positive.

Proof. Let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for V consisting eigenvectors of T. Let $C_1, \cdots, C_n$ be eigenvalues corresponding to $\alpha_1, \cdots, \alpha_n$, respectively. ⌋ setting

For each $1 \leq i \leq n$,

$$T^2(\alpha_i) = C_i^2 \cdot \alpha_i = U^2(\alpha_i).$$

And, $C_i^2 \alpha_i = U^2(\alpha_i) \Leftrightarrow (U^2 - C_i I)(\alpha_i) = 0.$

$$\Leftrightarrow (U + C_i I) \circ (U - C_i I)(\alpha_i) = 0.$$

If $V = (U - C_i I)(\alpha_i) \neq 0$, then

$$U(V) = -C_i V$$

So V is an eigenvector of U with negative eigenvalue.

It is contradiction to U is positive, so $U(\alpha_i) = C_i \alpha_i$ for all $1 \leq i \leq n$.

The linear map is determined by maps basis to some vectors, so $T = U$ ⌋(4)

Since T and U commute and both are Hermitian,

$$(TU)^* = U^* T^* = UT = TU$$

So TU is Hermitian. For the above $\mathcal{B}$, define a linear map as

$$\sqrt{T}(\alpha_i) = \sqrt{C_i} \cdot \alpha_i.$$

Then, for each $1 \leq i \leq n$,

$$U\sqrt{T}(\alpha_i) = U(\sqrt{C_i}\alpha_i) = \frac{1}{\sqrt{C_i}} \cdot U(C_i\alpha_i) = \frac{1}{\sqrt{C_i}} \cdot U(T(\alpha_i)) = \frac{1}{\sqrt{C_i}} \cdot TU(\alpha_i)$$

Def. A form f on an inn. prod space V is called non-negative if:

f is Hermitian and $f(\alpha,\alpha) \geq 0$ for all $\alpha \in V$.

and, if $f(\alpha,\alpha) > 0$, $\alpha \neq 0$, it is called positive.

Example. Let f be a form on fin. dim inn. prod space V, let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be an ordered basis.
Let A be a Matrix of f in $\mathcal{B}$. That is, $A_{jk} = f(\alpha_k, \alpha_j)$
If $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$, then

$$f(\alpha,\alpha) = f\left(\sum_j x_j\alpha_j, \sum_k x_k\alpha_k\right) = \sum_j \sum_k x_j \overline{x_k} \cdot f(\alpha_j, \alpha_k) = \sum_j \sum_k A_{kj} \cdot x_j \cdot \overline{x_k}.$$

So, f is non-negative if and only if $A = A^*$ and $\sum_j \sum_k A_{kj} x_j \overline{x_k} \geq 0$, for all x_j, x_k
Equivalently, f is non-negative if and only if

$$g : F^n \times F^n \to F : (X, Y) \mapsto Y^* A X$$

is non-negative form, $F = \mathbb{C}$ or $\mathbb{R}$.

Thm. Let $F = \mathbb{C}$ or $\mathbb{R}$, and $A \in M_{n\times n}(F)$.
A function $g : F^n \times F^n \to F : (X, Y) \mapsto Y^* A X$

is a positive form if and only if $A = P^*P$ for some invertible $P \in M_{n\times n}(F)$.

Proof Suppose that g is positive form. Then, it is an inner product on F^n,
So there is an orthonormal basis $\{e_1, \cdots, e_n\}$ for F^n, satisfying

$$g(e_i, e_j) = e_j^* A e_i = \delta_{ij}.$$

So, take $P = \begin{pmatrix} e_1 & \cdots & e_n \end{pmatrix}^{-1}$, then $PAP^* = I$, or $A = P^*P$.

Conversely, if $A = P^*P$ for some invertible P, then $A^* = (P^*P)^* = P^*P = A$ and

$$X^*AX = X^*P^*PX = (PX)^* \cdot (PX) > 0 \quad \text{if } X \neq 0.$$

So proved.

Def. Let $A \in M_{n\times n}(F)$. The principal minors of A are:

$$\Delta_k(A) \overset{def}{=} \det \begin{bmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \cdots & A_{kk} \end{bmatrix} \qquad 1 \le k \le n$$

Lemma. Let $A \in M_{n\times n}(F)$ be an invertible. TFAE:

a) There is a uni-upper triangle matrix P such that AP is lower triangular.

b) The principal minors of A are all different from 0.

Proof. Note: B is upper triangle iff $B_{jk} = 0$ if $j > k$.

Suppose that a) holds.

Denote $A_1, \cdots, A_n$ are columns of A, and $B_1, \cdots, B_n$ are column of $B = AP$.

Then, $B_1 = A_1$ by P is uni-upper, and

$$B_r = A_r + \sum_{j=1}^{r-1} P_{jr} \cdot A_j$$

(observe that

$$\begin{pmatrix} B_1 & \cdots & B_n \end{pmatrix} = \begin{pmatrix} A_1 & \cdots & A_n \end{pmatrix} \cdot \left(I + \begin{pmatrix} 0 & & P_{jk} \\ & 0 \ddots & \\ 0 & & 0 \end{pmatrix} \right) \qquad (P_{jk} = 0 \text{ if } j \ge k)$$

$$= \begin{pmatrix} A_1 & \cdots & A_n \end{pmatrix} + \Big(\underbrace{\sum_{j=1}^{r-1} P_{jr} \cdot A_j}_{r\text{'th column}} \Big)$$

).

Therefore, for each $1 \le k \le n$,

$\begin{bmatrix} B_{11} & \cdots & B_{1k} \\ \vdots & \ddots & \vdots \\ B_{k1} & \cdots & B_{kk} \end{bmatrix}$ is obtained by adding to the r th column of

$\begin{bmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \cdots & A_{kk} \end{bmatrix}$ as a linear combination of its other columns.

This operation cannot change the determinant, so, $\Delta_k(A) = \Delta_k(B)$.

And, since A, P are invertible, so is B, thus $\Delta_n(B) = \det B \ne 0$.

Now, $\Delta_k(B) = B_{11} \cdots B_{kk} \ne 0$, since it is triangular.